

Equilibrium

Resolving forces in two directions

YOUR AIM TODAY

Use zero resultant force to solve particle equilibrium.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

The ideas you need

A particle in equilibrium has no acceleration, so forces balance in every direction.

- Equilibrium means resultant force is zero.
- Resolve horizontally: $\sum F_x = 0$.
- Resolve vertically: $\sum F_y = 0$.
- A force P at angle θ has components $P \cos \theta$ horizontally and $P \sin \theta$ vertically.
- A string pulls away from the particle along the string.

MEMORY RULE

Draw arrows first; resolve second; solve last.

Two components

WORKED STEPS

A 10 N force at 30 degrees above horizontal has horizontal component $10\cos 30$ and vertical component $10\sin 30$.

Balance

WORKED STEPS

If tension T balances a 12 N horizontal force, the horizontal components must total 12 N in the opposite direction.

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Resolve 20 N at 60 degrees above horizontal.	
2. A particle has 8 N right and P left. Find P.	
3. Forces 6 N right and 8 N up are balanced by one force. Find its magnitude.	
4. State its direction as components.	
5. A weight 50 N is supported by two equal strings at 30 degrees above horizontal. Find each tension.	
6. Explain why equilibrium does not necessarily mean stationary.	

STOP

Do not continue until all six questions have an attempt.

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	Horizontal 10 N; vertical $10\sqrt{3}$ N.
2	$P=8$ N.
3	$\sqrt{6^2+8^2}=10$ N.
4	6 N left and 8 N down.
5	$2T \sin 30=50$, so $T=50$ N.
6	It can move with constant velocity because acceleration is zero.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.